RESEARCH NOTE · UK CLOSED-END FUNDS · 2007–2026

The Discount Machine

What nineteen years of free public data say about buying UK investment trusts below asset value — with full technical appendices.

Built from 1,003 archive files of the Association of Investment Companies and 4,632 dividend announcements from the Investegate regulatory news archive.
303 trusts, dead and alive · 41,241 fund-month observations · 235 months.

Historical backtests are research tools, not investment advice.
No missing financial data has been synthetically filled.

THE VERDICT IN THREE SENTENCES

Yes, buying UK investment trusts at unusually wide discounts has been genuinely profitable — the cheapest tenth of the market beat the average trust by roughly 7–15 percentage points a year before costs, a result too consistent to be luck.

But the profit is a sprint, not a marathon. Nearly all of it arrives in the first month after a fund becomes abnormally cheap; an investor who reacts a month late captures almost nothing, and since 2022 even the fast version has paid roughly zero.

And 15–20% a year from this alone is not supported. A disciplined, fast-moving investor could plausibly have earned low double digits before costs; the celebrated specialist returns must come from leverage, activism, or deal-level timing that monthly public data cannot capture.

What an investment trust is, and what we tested

An investment trust is a company whose only business is owning a portfolio of other investments. Because its own shares trade freely on the London Stock Exchange, the share price can drift away from the value of what it owns. When the shares sell for less than the portfolio per share, the gap is called the **discount** — paying 90p for £1.00 of assets is a 10% discount.

DISCOUNT & NAV
NAV — net asset value — is what one share's slice of the portfolio is worth. Discount = share price ÷ NAV − 1. Negative means cheap: −20% means 80p buys £1 of assets.

Specialist investors argue that buying at wide discounts and waiting for the gap to close is a durable source of profit. We tested that claim month by month from January 2007 to July 2026, using only what an investor could have known at the time, across 303 London-listed trusts including every one that later merged, liquidated or vanished. No missing number was invented; where data doesn't exist, the gap is reported, not filled.

How to read the numbers

CAGR
Compound annual growth rate — the steady yearly return producing the same end result. £1 at 10% CAGR for 19 years becomes about £6.12.

ALPHA
Return above what simply owning every trust would have earned — the part the tested pattern added.

T-STATISTIC
A luck detector: how large a result is relative to its noise. Above ~2, chance is an unlikely explanation; near 0, it could easily be luck.

Z-SCORE
How cheap a trust is versus its own normal. Always at −20% and there today: zero. Normally −5% and suddenly −20%: deeply negative — the dislocation the strategies hunt.

SKIP-MONTH TEST
Re-run everything ignoring the first month after each signal. Data errors and profits only a lightning-fast trader could catch disappear; what survives is slow enough for a real investor.

Primary returns are built from month-end share prices excluding dividends (no free source records dividends for long-dead funds). Dividend histories were separately rebuilt for most of the universe from company announcements, and headline results are shown on both bases.

The findings, in order of confidence

FINDING 1 · VERY HIGH CONFIDENCE

Cheap trusts really did beat expensive ones — in a near-perfect staircase

Sorted into ten buckets by discount each month, next-month returns descend almost perfectly from cheapest (+1.18%/month) to dearest (−0.45%). Strategies built on this earned 12.8–28.9% a year before costs against 5.1% for the average trust; with real dividends added back, 15.9% vs 7.9% on the covered sample. Ranking each trust against its own history (z-score) beat raw cheapness: ~14.7 points of annual alpha versus ~8.

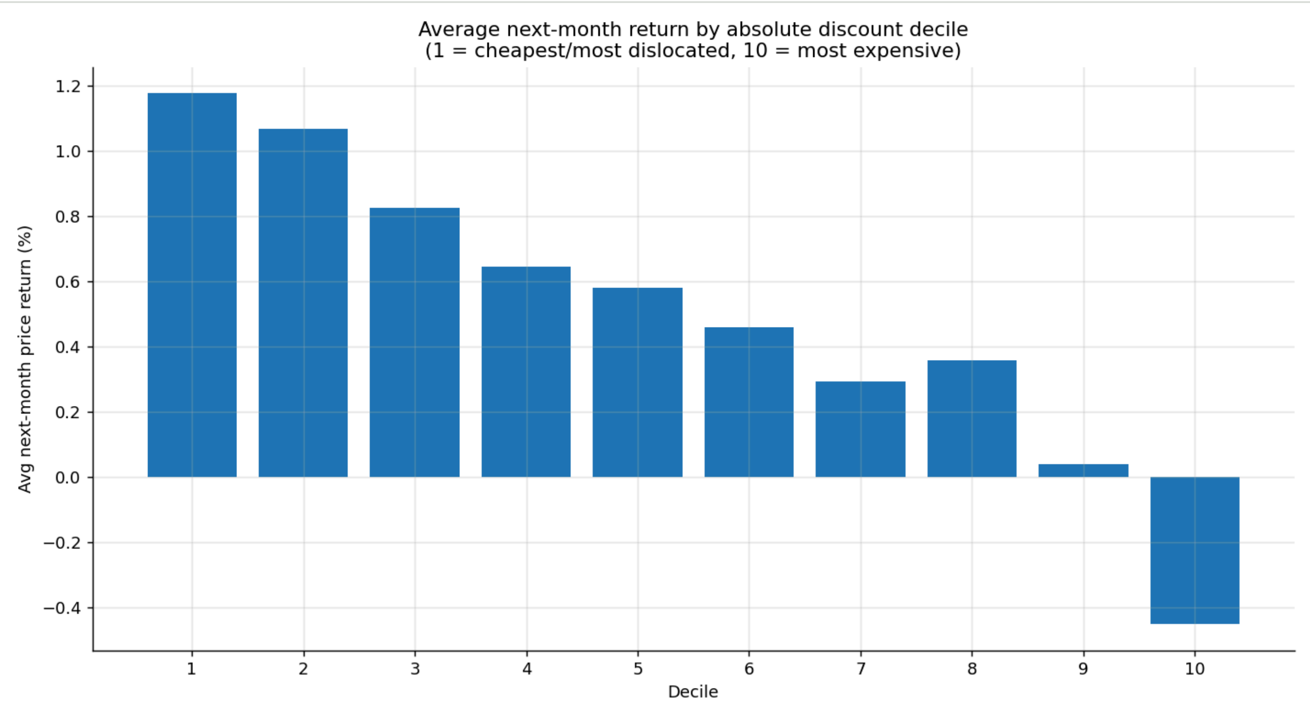

Fig. 1 — Average next-month return by discount decile, 2007–2026. Decile 1 = cheapest.

FINDING 2 · VERY HIGH CONFIDENCE

The profit is a sprint: skip one month and it nearly all evaporates

Under the skip-month test annual alpha collapses: 8.0→0.9 points (raw discount), 14.7→1.4 (z-score), 17.1→2.8 (composite). The snap-back happens within weeks of a trust becoming abnormally cheap. The discount machine pays the investor who monitors daily and acts at the turn of the month — and almost nothing to one who reads about the bargain later.

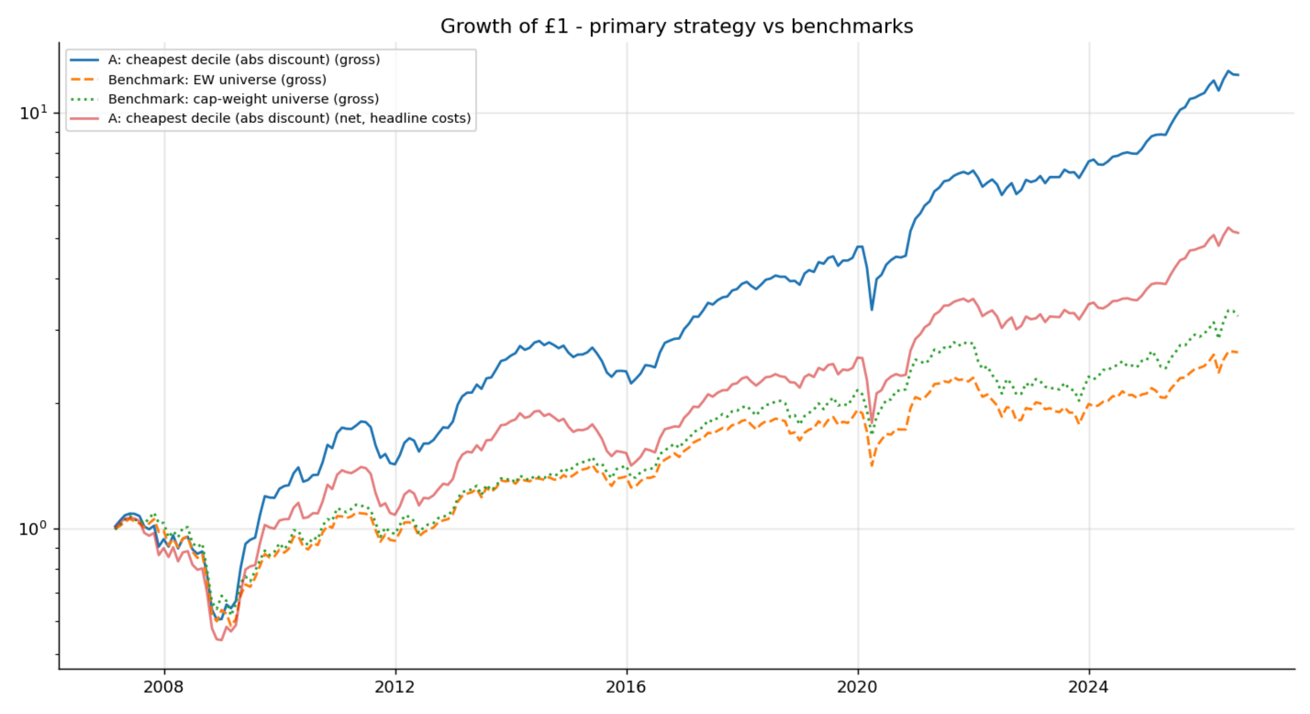

Fig. 2 — Growth of £1: cheapest-decile strategy vs whole-universe benchmark (price returns).

FINDING 3 · HIGH CONFIDENCE

Since 2022, the machine has stalled

Strong 2007–2016, present 2017–2021, roughly zero from 2022 onward — precisely the years UK trust discounts widened to a generation's extreme and stayed there. Cheap stopped snapping back. Whether today's discounts are a coiled spring or a permanent repricing is the judgment the data cannot make.

FINDING 4 · HIGH CONFIDENCE

The money came from the gap closing, not the assets growing

A trust's return decomposes exactly into portfolio growth (NAV), discount change, and dividends. Our best-quality portfolio's 15.3% annual return split as **5.6% NAV growth × 7.1% discount narrowing + 2.1% dividends**. Full year-by-year table: Appendix F.

FINDING 5 · MODERATE CONFIDENCE

"Catalysts" didn't help a monthly investor — the market moves first

Tested twice — industry records and real announcement dates from the regulatory news archive — cheap trusts with a recent tender/wind-up/merger announcement did no better next month (1.18% vs 2.08% monthly; weakly significant the wrong way). The discount re-rates on announcement day, before a month-end screen can react.

FINDING 6 · MODERATE CONFIDENCE, CORRECTED

"Great funds, temporarily cheap" helps — modestly, and only at speed

Buying only top-quartile five-year NAV compounders when wider than their own normal discount beat either ingredient alone — ~7 points of annual alpha in a 6–8 name portfolio — but the skip-month test flattens it to zero and it too has earned nothing since 2022.

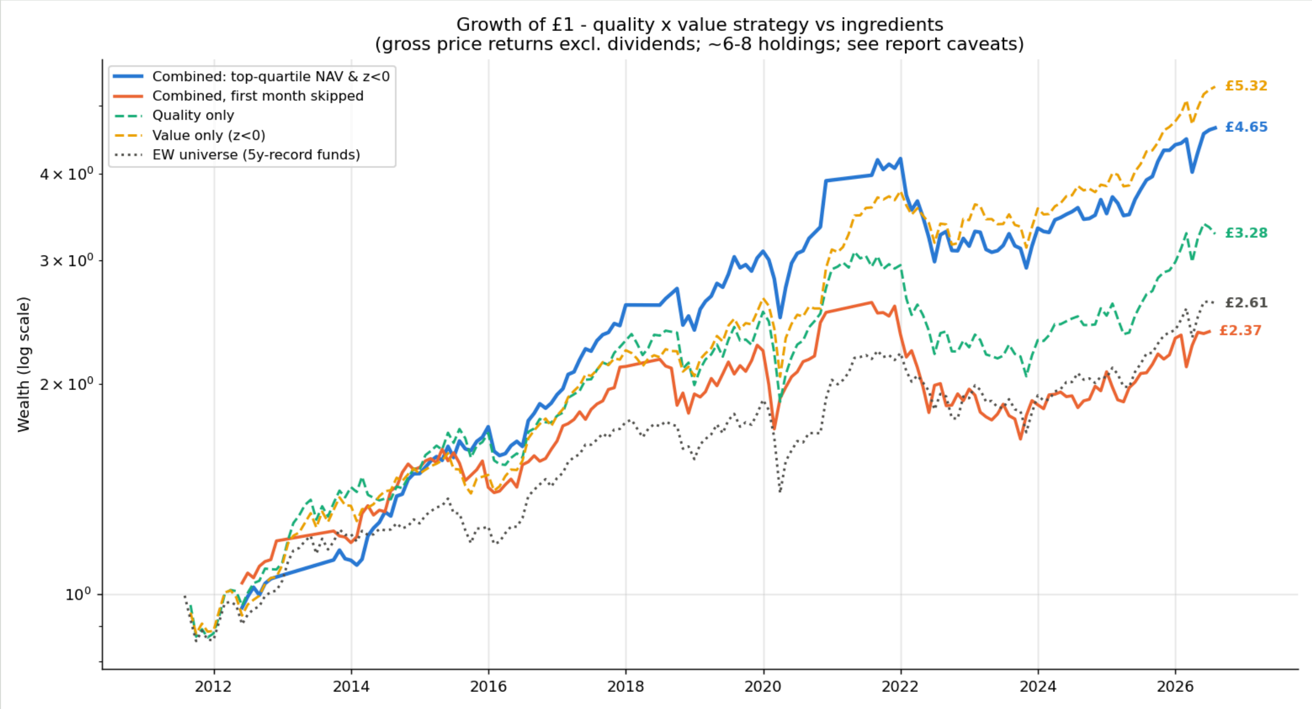

Fig. 3 — Quality × value vs its ingredients. The skip-month line hugging the benchmark is Finding 2 in one picture.

A CORRECTION, ON THE RECORD

An earlier draft showed this strategy earning 24–28% a year and surviving the skip-month test. That was a software bug: in months with fewer than five qualifiers the simulator re-counted the previous month's returns. Our own accounting identity — the decomposition refusing to add up — exposed it. A backtest you can't audit is a backtest you shouldn't trust, including ours.

FINDING 7 · CONTEXTUAL

In crashes it falls with everyone — then recovers roughly twice as fast

2008–09: cheapest decile −29% vs −39% for the average trust; the following twelve months +103% vs +58%. After COVID: +83% vs +50%. Mechanically, the strategy buys panic — painful to hold, well paid on the far side.

Costs, and the final verdict on 15–20%

At a realistic ½% each way plus stamp duty the cheapest-decile strategy falls from 13.7% to ~9.1% a year; at 1% each way, to 6.9% — barely ahead of the 5.1% benchmark (Appendix I). The honest ledger: the average trust returned ~5% in price terms (~8% with dividends); systematic discount buying added mid-single to low-double-digit alpha before costs for an investor fast enough to act within the month; costs claim a third or more; the engine has idled since 2022. A specialist earning 15–20% must be adding leverage, activist pressure, announcement-day dealing, or genuine selection inside portfolios. The discount is real fuel; it is not the whole engine.

APPENDICES

All tables regenerate from the repository (outputs/*.csv); every number traces to named AIC or Investegate source files via data/manifest.csv.

APPENDIX A · Data & methodology

Universe. Point-in-time monthly membership from the AIC Monthly Information Release (MIR) CSV archive, Jan 2007–Jul 2026: ordinary shares of conventional London-listed investment companies; VCTs, split-capital classes, ZDPs and non-sterling quote lines excluded. Dead, merged and renamed trusts included through their final published month; entities keyed by SEDOL (UK ISINs embed the SEDOL, chaining identifier eras), renames linked via corporate-activity records.

Returns. Month-end MIR mid prices; splits/consolidations adjusted from shares-in-issue; pence/pounds unit switches corrected only where a ×100 rescale restores price/NAV consistency (26 rows) and returns invalidated — never repaired — where the implied one-month discount move exceeds 4× without NAV corroboration (33 rows). Dividends parsed from Investegate announcement pages (ex-date attach; 69% of 4,632 with exact ex-dates), validated per security-year against the AIC's independently published trailing yield; failing years excluded from total-return eligibility, never silently understated.

Timing. Month-end t information trades in month t+1 (AIC publishes ~6 working days after month-end); late-reported rows are never signal-eligible before publication. Skip-month variants earn month t+2. Development 2007–16 / validation 2017–21 / holdout 2022+ splits were fixed before results were inspected; the overshoot composite's weights (50/25/25) and z-window (36m) were pre-specified.

Engine safeguards. Missing returns never become 0 or −100%; delistings classified against corporate-activity outcomes and left unresolved where the payoff is unknowable; portfolio months with insufficient qualifiers hold the prior book at current-month prices (the corrected carry-forward); 74 automated tests including parser regressions on real archive files.

Appendix B1 · Performance summary — full period, all strategies and bases

STRATEGY	PERIOD	BASIS	MO	CAGR	VOL	SHARPE	MAXDD	ALPHA	T
A_absolute_discount_decile	full	gross	234	13.7%	16.2%	0.88	-44.4%	8.0%	4.79
A_absolute_discount_decile	full	net_headline	234	8.8%	16.2%	0.60	-49.2%	3.5%	2.10
B_absolute_discount_quintile	full	gross	234	13.0%	15.6%	0.86	-45.1%	7.1%	6.68
B_absolute_discount_quintile	full	net_headline	234	8.3%	15.6%	0.59	-48.8%	2.8%	2.66
C_discount_zscore	full	gross	211	25.6%	14.6%	1.65	-17.7%	14.7%	10.39
C_discount_zscore	full	net_headline	211	14.3%	14.4%	1.00	-19.6%	5.2%	3.74
D_sector_neutral_zscore	full	gross	211	23.2%	14.0%	1.57	-20.6%	12.6%	13.06
D_sector_neutral_zscore	full	net_headline	211	12.2%	13.9%	0.90	-22.8%	3.1%	3.41
E_discount_overshoot	full	gross	211	28.9%	15.0%	1.78	-20.0%	17.1%	11.92
E_discount_overshoot	full	net_headline	211	16.1%	14.9%	1.08	-21.9%	6.5%	4.62
BM_equal_weight_universe	full	gross	234	5.1%	13.9%	0.43	-44.7%	–	–
BM_equal_weight_universe	full	net_headline	234	4.5%	13.9%	0.39	-45.5%	–	–
BM_cap_weight_universe	full	gross	234	6.2%	14.4%	0.49	-43.5%	1.0%	1.35
BM_cap_weight_universe	full	net_headline	234	5.5%	14.5%	0.45	-44.6%	0.4%	0.48
BM_equal_weight_universe_TR	full	gross_TR	234	7.9%	14.1%	0.61	-40.7%	–	–
A_absolute_discount_decile_TR	full	gross_TR	234	15.9%	17.6%	0.93	-37.9%	7.2%	3.41
C_discount_zscore_TR	full	gross_TR	211	26.1%	15.1%	1.62	-18.6%	12.8%	7.37
E_discount_overshoot_TR	full	gross_TR	211	30.0%	15.8%	1.76	-21.1%	15.6%	8.41
F_quality_only	full	gross	180	8.2%	14.4%	0.62	-33.6%	1.5%	0.98
BM_5y_record_universe	full	gross	181	6.6%	12.8%	0.56	-26.5%	–	–
F_value_only	full	gross	180	11.8%	12.9%	0.93	-22.8%	4.8%	7.35
F_combined_z0.0	full	gross	150	13.1%	14.7%	0.91	-30.3%	6.8%	3.19
F_combined_z-0.5	full	gross	120	13.4%	16.0%	0.87	-26.7%	6.8%	2.05
F_combined_z-1.0	full	gross	70	11.9%	14.2%	0.87	-18.6%	7.5%	2.03
F_quality_only_TR	full	gross_TR	180	10.3%	14.6%	0.75	-30.0%	0.7%	0.43
BM_5y_record_universe	full	gross_TR	181	9.4%	12.9%	0.76	-26.4%	–	–
F_value_only_TR	full	gross_TR	180	14.5%	13.0%	1.11	-23.2%	4.6%	6.92
F_combined_z0.0_TR	full	gross_TR	150	14.7%	15.2%	0.98	-27.9%	5.2%	2.43
F_combined_z-0.5_TR	full	gross_TR	117	15.9%	16.1%	1.00	-24.0%	6.6%	1.94
F_combined_z-1.0_TR	full	gross_TR	64	14.8%	13.8%	1.07	-15.2%	5.4%	1.35
F_quality_only_SKIP1M	full	gross_skip1m	179	8.0%	14.7%	0.60	-36.6%	0.5%	0.31
F_value_only_SKIP1M	full	gross_skip1m	179	8.2%	12.9%	0.68	-24.1%	1.0%	1.38
F_combined_z0.0_SKIP1M	full	gross_skip1m	149	7.2%	14.8%	0.55	-36.3%	-0.0%	-0.00
F_combined_z-0.5_SKIP1M	full	gross_skip1m	119	8.7%	15.2%	0.62	-34.1%	-1.4%	-0.51
F_combined_z-1.0_SKIP1M	full	gross_skip1m	70	8.9%	16.5%	0.60	-22.7%	-1.4%	-0.33

Appendix B2 · Development / validation / holdout sub-periods (gross)

STRATEGY	PERIOD	BASIS	MO	CAGR	VOL	SHARPE	MAXDD	ALPHA	T
A_absolute_discount_decile	development	gross	119	11.8%	17.8%	0.72	-44.4%	6.9%	2.57
A_absolute_discount_decile	validation	gross	60	19.2%	17.2%	1.12	-29.5%	8.8%	3.37
A_absolute_discount_decile	holdout	gross	55	12.3%	10.6%	1.14	-9.1%	9.3%	3.67
B_absolute_discount_quintile	development	gross	119	11.3%	16.9%	0.72	-45.1%	6.2%	3.73
B_absolute_discount_quintile	validation	gross	60	18.2%	15.9%	1.14	-26.8%	8.2%	4.99
B_absolute_discount_quintile	holdout	gross	55	11.0%	12.0%	0.93	-12.5%	7.7%	4.09
C_discount_zscore	development	gross	96	29.0%	14.7%	1.82	-12.9%	13.6%	6.60
C_discount_zscore	validation	gross	60	27.2%	15.6%	1.63	-17.7%	16.3%	5.49
C_discount_zscore	holdout	gross	55	18.3%	13.2%	1.35	-10.8%	14.0%	5.66
D_sector_neutral_zscore	development	gross	96	28.8%	14.0%	1.89	-12.4%	13.5%	9.80
D_sector_neutral_zscore	validation	gross	60	24.5%	15.1%	1.54	-20.6%	13.9%	7.12
D_sector_neutral_zscore	holdout	gross	55	12.7%	12.6%	1.01	-10.3%	9.0%	5.25
E_discount_overshoot	development	gross	96	33.2%	15.5%	1.95	-14.5%	16.3%	7.45
E_discount_overshoot	validation	gross	60	27.6%	16.0%	1.62	-20.0%	16.3%	6.04
E_discount_overshoot	holdout	gross	55	23.2%	13.2%	1.66	-7.0%	18.1%	6.99
BM_equal_weight_universe	development	gross	119	4.4%	14.5%	0.37	-44.7%	–	–
BM_equal_weight_universe	validation	gross	60	8.4%	14.3%	0.64	-26.5%	–	–
BM_equal_weight_universe	holdout	gross	55	3.1%	12.2%	0.31	-17.9%	–	–
BM_cap_weight_universe	development	gross	119	5.0%	14.9%	0.40	-43.5%	0.5%	0.62
BM_cap_weight_universe	validation	gross	60	11.5%	13.6%	0.87	-22.2%	3.4%	2.28
BM_cap_weight_universe	holdout	gross	55	3.3%	14.6%	0.30	-19.7%	-0.1%	-0.04

Splits fixed ex-ante: 2007–16 / 2017–21 / 2022+.

Appendix C · Decile tests — all four signals

SIGNAL	BUCKET	AVG FWD/MO	T	SHARPE	WIN	MO
absolute_discount	1	1.18%	3.86	0.87	67.9%	234
absolute_discount	2	1.07%	3.52	0.80	66.2%	234
absolute_discount	3	0.83%	2.79	0.63	63.7%	234
absolute_discount	4	0.64%	2.22	0.50	57.7%	234
absolute_discount	5	0.58%	2.06	0.47	60.3%	234
absolute_discount	6	0.46%	1.63	0.37	60.7%	234
absolute_discount	7	0.29%	1.09	0.25	56.0%	234
absolute_discount	8	0.36%	1.47	0.33	58.1%	234
absolute_discount	9	0.04%	0.15	0.03	56.4%	234
absolute_discount	10	-0.45%	-1.97	-0.45	47.4%	234
absolute_discount	1-10 (long-short)	1.63%	8.98	2.03	73.9%	234
discount_zscore	1	1.99%	6.85	1.63	73.0%	211
discount_zscore	2	1.33%	5.01	1.19	66.4%	211
discount_zscore	3	1.08%	4.25	1.01	64.5%	211
discount_zscore	4	0.90%	3.33	0.79	64.5%	211
discount_zscore	5	0.80%	2.98	0.71	61.6%	211
discount_zscore	6	0.77%	2.88	0.69	59.7%	211
discount_zscore	7	0.61%	2.32	0.55	59.2%	211
discount_zscore	8	0.45%	1.67	0.40	58.8%	211
discount_zscore	9	0.14%	0.53	0.13	52.6%	211
discount_zscore	10	-0.50%	-1.83	-0.44	50.2%	211
discount_zscore	1-10 (long-short)	2.50%	12.50	2.98	84.8%	211
widening_3m	1	1.77%	5.92	1.35	71.9%	231
widening_3m	2	1.11%	3.90	0.89	64.9%	231
widening_3m	3	0.93%	3.41	0.78	61.9%	231
widening_3m	4	0.73%	2.69	0.61	62.3%	231
widening_3m	5	0.60%	2.25	0.51	61.5%	231
widening_3m	6	0.50%	1.87	0.43	58.4%	231
widening_3m	7	0.38%	1.38	0.31	57.6%	231
widening_3m	8	0.07%	0.25	0.06	54.5%	231
widening_3m	9	-0.09%	-0.30	-0.07	51.9%	231
widening_3m	10	-1.04%	-3.60	-0.82	40.7%	231
widening_3m	1-10 (long-short)	2.81%	13.86	3.16	86.6%	231
overshoot	1	2.21%	7.41	1.77	75.8%	211
overshoot	2	1.43%	5.16	1.23	66.8%	211
overshoot	3	1.18%	4.79	1.14	67.3%	211
overshoot	4	1.03%	3.76	0.90	67.8%	211
overshoot	5	0.73%	2.70	0.64	61.1%	211
overshoot	6	0.73%	2.73	0.65	62.1%	211
overshoot	7	0.51%	1.97	0.47	58.3%	211
overshoot	8	0.29%	1.11	0.26	55.0%	211
overshoot	9	0.04%	0.16	0.04	53.1%	211
overshoot	10	-0.56%	-2.12	-0.51	47.9%	211
overshoot	1-10 (long-short)	2.76%	14.55	3.47	87.2%	211

Bucket 1 = cheapest/most dislocated. Long-short = bucket 1 minus 10.

Appendix D · Robustness grid incl. skip-month variants

VARIANT	MO	CAGR	SHARPE	ALPHA	T	AVG HOLD	MAXDD
discount_top5	235	12.6%	0.77	7.4%	3.12	9	-35.9%
discount_top10	235	13.7%	0.88	8.0%	4.79	18	-44.4%
discount_top20	235	13.0%	0.86	7.1%	6.68	35	-45.1%
discount_top30	235	11.7%	0.80	5.8%	7.77	53	-43.0%
z24m_top10	212	27.5%	1.79	16.5%	11.08	16	-14.0%
z36m_top10	212	25.6%	1.65	14.7%	10.39	17	-17.7%
z60m_top10	212	22.5%	1.49	12.1%	8.93	17	-22.4%
z36m_monthly	212	25.6%	1.65	14.7%	10.39	17	-17.7%
z36m_quarterly	212	13.9%	0.98	4.8%	3.54	16	-26.0%
z36m_equal	212	25.6%	1.65	14.7%	10.39	17	-17.7%
z36m_market_cap	212	21.0%	1.30	10.7%	5.84	17	-18.9%
z36m_capped_inverse_rank	212	29.4%	1.78	17.6%	10.37	17	-16.0%
z36m_mcap0	212	25.6%	1.65	14.7%	10.39	17	-17.7%
z36m_mcap25	212	24.5%	1.54	13.4%	9.96	16	-18.5%
z36m_mcap50	212	22.3%	1.43	11.7%	8.52	14	-18.4%
z36m_mcap100	212	21.4%	1.36	11.1%	6.96	12	-22.3%
z36m_mcap200	212	18.7%	1.24	9.2%	5.41	9	-21.9%
discount_top10_SKIP1M	235	6.0%	0.44	0.9%	0.51	18	-53.3%
z36m_top10_SKIP1M	212	10.3%	0.77	1.4%	1.17	17	-27.3%
overshoot_top10_SKIP1M	212	12.0%	0.86	2.8%	2.28	17	-28.8%
z36m_mcap100_SKIP1M	212	8.9%	0.65	-0.3%	-0.23	12	-26.6%
overshoot_mcap100_SKIP1M	212	11.1%	0.78	1.6%	1.26	12	-26.5%

Appendix E · Quality × value 4×4 double-sort (t+1 and skip-month)

HORIZON	NAV Q	Z Q	AVG FWD/MO	T	MO	NAMES/MO
t+1	1	1	1.42%	4.00	168	3
t+1	1	2	0.89%	2.37	179	3
t+1	1	3	0.84%	2.29	176	4
t+1	1	4	0.27%	0.76	178	5
t+1	2	1	1.09%	3.55	174	4
t+1	2	2	0.74%	2.44	180	4
t+1	2	3	0.52%	1.77	174	4
t+1	2	4	-0.01%	-0.03	178	4
t+1	3	1	1.53%	5.16	177	4
t+1	3	2	0.65%	2.25	178	4
t+1	3	3	0.37%	1.30	180	4
t+1	3	4	-0.11%	-0.33	176	4
t+1	4	1	0.98%	2.73	176	4
t+1	4	2	0.80%	2.17	180	4
t+1	4	3	0.20%	0.57	177	4
t+1	4	4	-0.34%	-1.05	174	4
t+2_skip	1	1	0.88%	2.24	166	3
t+2_skip	1	2	0.92%	2.29	177	3
t+2_skip	1	3	0.74%	1.99	175	4
t+2_skip	1	4	0.69%	1.80	177	5
t+2_skip	2	1	0.66%	2.25	173	4
t+2_skip	2	2	0.49%	1.57	179	4
t+2_skip	2	3	0.74%	2.39	173	4
t+2_skip	2	4	0.47%	1.51	177	4
t+2_skip	3	1	0.74%	2.55	176	4
t+2_skip	3	2	0.73%	2.54	176	4
t+2_skip	3	3	0.39%	1.33	179	4
t+2_skip	3	4	0.95%	2.95	175	4
t+2_skip	4	1	0.83%	2.49	174	4
t+2_skip	4	2	0.59%	1.55	179	4
t+2_skip	4	3	0.73%	1.88	176	4
t+2_skip	4	4	-0.10%	-0.31	173	4

NAV Q1 = best 5y compounders; Z Q1 = most dislocated vs own history.

Appendix F · Annual return decomposition — quality × value strategy

YEAR	TOTAL	NAV GROWTH	DISCOUNT	DIVIDENDS	RESIDUAL	MO	COVERAGE
2012.0	5.8%	4.0%	1.8%	0.0%	-0.0%	7	100.0%
2013.0	5.6%	1.1%	4.6%	0.0%	0.0%	4	100.0%
2014.0	33.3%	27.7%	4.0%	0.1%	1.4%	12	100.0%
2015.0	17.6%	10.2%	4.8%	2.0%	0.7%	12	97.4%
2016.0	12.5%	8.5%	2.5%	1.2%	0.3%	12	100.0%
2017.0	36.1%	16.9%	15.1%	1.3%	2.8%	12	100.0%
2018.0	-6.2%	-13.1%	6.0%	1.8%	-0.9%	7	100.0%
2019.0	35.5%	21.0%	7.0%	4.5%	2.9%	12	100.0%
2020.0	31.9%	14.8%	9.7%	4.7%	2.7%	11	100.0%
2021.0	8.5%	1.4%	6.1%	0.9%	0.1%	6	100.0%
2022.0	-23.9%	-25.9%	1.0%	1.6%	-0.5%	12	98.8%
2023.0	8.3%	6.1%	0.2%	2.0%	-0.1%	12	99.5%
2024.0	6.3%	-2.6%	8.5%	1.3%	-0.9%	12	100.0%
2025.0	29.7%	12.0%	13.1%	3.4%	1.3%	12	99.0%
2026.0	6.7%	0.4%	5.6%	1.0%	-0.2%	7	100.0%

(1+total) = (1+NAV)(1+discount) + dividends; residual = compounding cross-terms.

Appendix G · Fama-MacBeth cross-sectional regressions

SPECIFICATION	VARIABLE	MEAN COEF	T (NEWEY-WEST)	MO
univariate_discount	const	0.0016	0.60	234
univariate_discount	discount	-0.0416	-5.37	234
univariate_zscore	const	0.0069	2.68	211
univariate_zscore	discount_z_36m	-0.0059	-10.53	211
univariate_widening3m	const	0.0044	1.46	231
univariate_widening3m	discount_change_3m	-0.1831	-12.38	231
multivariate	const	-0.0001	-0.03	211
multivariate	discount	-0.0241	-3.63	211
multivariate	discount_z_36m	-0.0028	-5.75	211
multivariate	discount_change_3m	-0.1490	-9.90	211
multivariate	log_mcap	0.0010	3.03	211
multivariate_sector_demeaned	const	-0.0000	-0.03	211
multivariate_sector_demeaned	discount	-0.0224	-3.82	211
multivariate_sector_demeaned	discount_z_36m	-0.0029	-6.66	211
multivariate_sector_demeaned	discount_change_3m	-0.1661	-12.15	211
multivariate_sector_demeaned	log_mcap	0.0006	2.25	211

Monthly cross-sections; NW(4) t-stats on coefficient time series. No causal claims.

Appendix H · Market-stress episodes and +12-month recoveries

EPISODE	STRATEGY	MO	CUM. RETURN	WORST MO
GFC 2008-09	A_absolute_discount_decile	15	-29.0%	-17.5%
GFC 2008-09	B_absolute_discount_quintile	15	-34.0%	-19.1%
GFC 2008-09	C_discount_zscore	3	9.4%	-5.3%
GFC 2008-09	D_sector_neutral_zscore	3	4.1%	-2.6%
GFC 2008-09	E_discount_overshoot	3	5.7%	-6.2%
GFC 2008-09	BM_equal_weight_universe	15	-38.7%	-17.3%
GFC 2008-09	BM_cap_weight_universe	15	-36.2%	-16.4%
GFC 2008-09 +12m recovery	A_absolute_discount_decile	12	102.9%	-0.8%
GFC 2008-09 +12m recovery	BM_equal_weight_universe	12	57.7%	-2.0%
Eurozone crisis	A_absolute_discount_decile	14	-9.5%	-9.6%
Eurozone crisis	B_absolute_discount_quintile	14	-10.1%	-9.5%
Eurozone crisis	C_discount_zscore	14	10.0%	-6.9%
Eurozone crisis	D_sector_neutral_zscore	14	8.1%	-6.8%
Eurozone crisis	E_discount_overshoot	14	5.0%	-7.3%
Eurozone crisis	BM_equal_weight_universe	14	-10.3%	-7.3%
Eurozone crisis	BM_cap_weight_universe	14	-11.7%	-7.6%
Eurozone crisis +12m recovery	A_absolute_discount_decile	12	35.3%	-2.3%
Eurozone crisis +12m recovery	BM_equal_weight_universe	12	21.4%	-4.6%
2015-16 selloff	A_absolute_discount_decile	9	-15.9%	-6.4%
2015-16 selloff	B_absolute_discount_quintile	9	-11.1%	-6.1%
2015-16 selloff	C_discount_zscore	9	-8.8%	-5.3%
2015-16 selloff	D_sector_neutral_zscore	9	-5.2%	-6.1%
2015-16 selloff	E_discount_overshoot	9	-3.6%	-7.2%
2015-16 selloff	BM_equal_weight_universe	9	-11.3%	-5.9%
2015-16 selloff	BM_cap_weight_universe	9	-10.7%	-6.2%
2015-16 selloff +12m recovery	A_absolute_discount_decile	12	41.1%	-0.8%
2015-16 selloff +12m recovery	BM_equal_weight_universe	12	27.6%	-2.0%
Q4 2018	A_absolute_discount_decile	3	-4.4%	-2.4%
Q4 2018	B_absolute_discount_quintile	3	-5.6%	-3.8%
Q4 2018	C_discount_zscore	3	-1.0%	-2.8%
Q4 2018	D_sector_neutral_zscore	3	-7.1%	-5.2%
Q4 2018	E_discount_overshoot	3	-3.1%	-3.8%
Q4 2018	BM_equal_weight_universe	3	-10.1%	-6.6%
Q4 2018	BM_cap_weight_universe	3	-11.3%	-7.5%
Q4 2018 +12m recovery	A_absolute_discount_decile	12	23.4%	-5.1%
Q4 2018 +12m recovery	BM_equal_weight_universe	12	18.4%	-4.1%
COVID crash	A_absolute_discount_decile	2	-29.5%	-20.9%
COVID crash	B_absolute_discount_quintile	2	-26.1%	-18.3%
COVID crash	C_discount_zscore	2	-17.7%	-9.9%
COVID crash	D_sector_neutral_zscore	2	-20.6%	-11.7%
COVID crash	E_discount_overshoot	2	-20.0%	-10.9%
COVID crash	BM_equal_weight_universe	2	-25.1%	-17.2%
COVID crash	BM_cap_weight_universe	2	-20.4%	-12.9%
COVID crash +12m recovery	A_absolute_discount_decile	12	82.4%	-0.4%
COVID crash +12m recovery	BM_equal_weight_universe	12	50.1%	-1.1%
2022 rate shock	A_absolute_discount_decile	12	-6.2%	-5.9%
2022 rate shock	B_absolute_discount_quintile	12	-5.5%	-6.5%
2022 rate shock	C_discount_zscore	12	9.1%	-6.1%
2022 rate shock	D_sector_neutral_zscore	12	1.0%	-6.7%
2022 rate shock	E_discount_overshoot	12	14.7%	-5.5%
2022 rate shock	BM_equal_weight_universe	12	-16.0%	-6.6%
2022 rate shock	BM_cap_weight_universe	12	-21.6%	-9.5%
2022 rate shock +12m recovery	A_absolute_discount_decile	12	12.1%	-3.9%

EPISODE	STRATEGY	MO	CUM. RETURN	WORST MO
2022 rate shock +12m recovery	BM_equal_weight_universe	12	3.1%	-6.1%
2023-25 IT discount crisis	A_absolute_discount_decile	36	63.9%	-3.9%
2023-25 IT discount crisis	B_absolute_discount_quintile	36	56.5%	-4.5%
2023-25 IT discount crisis	C_discount_zscore	36	70.4%	-5.5%
2023-25 IT discount crisis	D_sector_neutral_zscore	36	47.5%	-5.7%
2023-25 IT discount crisis	E_discount_overshoot	36	95.7%	-5.6%
2023-25 IT discount crisis	BM_equal_weight_universe	36	27.2%	-6.1%
2023-25 IT discount crisis	BM_cap_weight_universe	36	34.4%	-5.6%
2023-25 IT discount crisis +12m recovery	A_absolute_discount_decile	8	10.5%	-5.4%
2023-25 IT discount crisis +12m recovery	BM_equal_weight_universe	8	7.8%	-9.4%

Appendix I · Transaction-cost scenarios

STRATEGY	ONE-WAY BPS	DUTY BPS	NET CAGR	NET SHARPE
A_absolute_discount_decile	0	0	13.7%	0.88
A_absolute_discount_decile	0	50	12.2%	0.79
A_absolute_discount_decile	25	0	12.0%	0.79
A_absolute_discount_decile	25	50	10.4%	0.70
A_absolute_discount_decile	50	0	10.3%	0.69
A_absolute_discount_decile	50	50	8.8%	0.60
A_absolute_discount_decile	100	0	6.9%	0.50
A_absolute_discount_decile	100	50	5.4%	0.41
B_absolute_discount_quintile	0	0	13.0%	0.86
B_absolute_discount_quintile	0	50	11.5%	0.78
B_absolute_discount_quintile	25	0	11.4%	0.77
B_absolute_discount_quintile	25	50	9.9%	0.69
B_absolute_discount_quintile	50	0	9.8%	0.68
B_absolute_discount_quintile	50	50	8.3%	0.59
B_absolute_discount_quintile	100	0	6.7%	0.49
B_absolute_discount_quintile	100	50	5.2%	0.41
C_discount_zscore	0	0	25.6%	1.65
C_discount_zscore	0	50	21.9%	1.45
C_discount_zscore	25	0	21.6%	1.43
C_discount_zscore	25	50	18.1%	1.23
C_discount_zscore	50	0	17.8%	1.21
C_discount_zscore	50	50	14.3%	1.00
C_discount_zscore	100	0	10.4%	0.76
C_discount_zscore	100	50	7.2%	0.55
D_sector_neutral_zscore	0	0	23.2%	1.57
D_sector_neutral_zscore	0	50	19.6%	1.36
D_sector_neutral_zscore	25	0	19.3%	1.34
D_sector_neutral_zscore	25	50	15.8%	1.13
D_sector_neutral_zscore	50	0	15.6%	1.12
D_sector_neutral_zscore	50	50	12.2%	0.90
D_sector_neutral_zscore	100	0	8.5%	0.66
D_sector_neutral_zscore	100	50	5.3%	0.44
E_discount_overshoot	0	0	28.9%	1.78
E_discount_overshoot	0	50	24.7%	1.56
E_discount_overshoot	25	0	24.4%	1.54
E_discount_overshoot	25	50	20.4%	1.32
E_discount_overshoot	50	0	20.1%	1.31
E_discount_overshoot	50	50	16.1%	1.08
E_discount_overshoot	100	0	11.8%	0.82
E_discount_overshoot	100	50	8.1%	0.60

Appendix J1 · Catalysts — AIC completion-month proxy

GROUP	N	MEAN FWD/MO	MEDIAN	T
cheap_no_catalyst	4481	2.06%	1.65%	21.15
cheap_with_catalyst	104	1.81%	1.24%	2.55
difference (catalyst - none)	4563	-0.25%	72.32%	-0.36

Appendix J2 · Catalysts — real announcement dates (Investegate)

GROUP	N	MEAN FWD/MO	MEDIAN	T
cheap_no_announced_catalyst	4452	2.08%	1.66%	21.25
cheap_with_announced_catalyst	133	1.18%	0.89%	2.05
difference (announced - none)	4563	-0.91%	12.09%	-1.56

Appendix K1 · Twenty best holding episodes — quality × value strategy

TRUST	ENTRY	HELD	ENTRY DISC	ENTRY Z	SPELL RETURN	CONTRIBUTION
JPMorgan European Smaller Companies	2020-06	6	-18.0%	-1.76	38.7%	11.7%
Henderson Smaller Companies	2017-07	6	-14.2%	-0.30	15.7%	8.7%
Biotech Growth	2014-05	7	-4.6%	-0.41	58.2%	8.5%
JPMorgan American	2018-12	8	-5.2%	-0.75	11.0%	7.6%
JPMorgan American	2020-06	6	-5.9%	-1.33	21.3%	7.2%
Aberdeen New Thai	2013-09	10	-17.3%	-0.91	7.5%	6.0%
Standard Life UK Smaller Companies	2013-09	4	-4.0%	-0.70	14.4%	5.8%
International Biotechnology	2014-06	5	-17.7%	-1.70	39.0%	5.7%
Edinburgh Dragon	2012-05	7	-9.5%	-0.87	5.8%	5.7%
Aberdeen Smaller Companies High Income	2014-08	38	-12.1%	-0.12	34.7%	5.4%
Aberdeen Latin American Income	2018-11	3	-13.3%	-0.28	8.2%	5.1%
JPMorgan Chinese	2019-06	4	-14.0%	-0.49	20.0%	4.9%
Schroder Asian Total Return	2020-02	9	-2.6%	-1.54	26.0%	4.6%
Henderson Smaller Companies	2016-04	14	-16.0%	-1.16	32.7%	4.2%
Hansa Investment Company	2026-02	4	-44.7%	-1.28	22.6%	4.0%
BlackRock North American Income	2020-03	2	-2.8%	-0.23	-3.3%	3.3%
JPMorgan Global Growth & Income	2025-03	18	-0.9%	-1.03	5.4%	3.2%
Fidelity China Special Situations	2016-05	6	-15.8%	-1.14	37.1%	3.0%
Geiger Counter	2025-09	4	-16.9%	-0.25	19.8%	3.0%
Fidelity Japan	2020-05	1	-12.9%	-1.15	19.7%	2.8%

Appendix K2 · Ten worst holding episodes

TRUST	ENTRY	HELD	ENTRY DISC	ENTRY Z	SPELL RETURN	CONTRIBUTION
Edinburgh Worldwide	2021-08	16	0.2%	-0.51	-46.0%	-6.4%
Geiger Counter	2026-02	4	-17.6%	-0.23	-16.3%	-4.2%
JPMorgan UK Smaller Companies	2022-01	15	-11.0%	-0.27	-35.0%	-4.2%
JPMorgan China Growth & Income	2023-03	11	-4.6%	-0.13	-43.5%	-4.0%
JPMorgan European Smaller Companies	2020-02	2	-11.9%	-0.14	-34.1%	-3.5%
TR European Growth	2018-06	6	-11.2%	-0.31	-16.7%	-3.5%
Montanaro European Smaller Companies	2021-07	15	-3.0%	-0.17	-37.0%	-3.0%
JPMorgan Chinese	2018-09	2	-14.5%	-0.37	-14.6%	-2.5%
abrdn UK Smaller Companies Growth	2021-11	6	-6.5%	-0.09	-22.4%	-2.4%
TR European Growth	2019-11	4	-13.7%	-0.83	-2.3%	-2.1%

Appendix K3 · Top contributors (share of strategy profit)

TRUST	TOTAL CONTRIBUTION	EPISODES	MONTHS HELD
JPMorgan American	18.6%	13	35
Biotech Growth	13.8%	8	45
Standard Life UK Smaller Companies	13.1%	12	43
Henderson Smaller Companies	12.9%	4	24
International Biotechnology	9.8%	8	13
JPMorgan European Smaller Companies	8.2%	2	8
JPMorgan Global Growth & Income	7.5%	8	46
Aberdeen New Thai	7.2%	6	21
Aberdeen Smaller Companies High Income	6.4%	2	39
Edinburgh Dragon	5.7%	1	7
Hansa Investment Company	5.4%	2	5
Fidelity Japan	5.3%	3	3
F&C Investment Trust	5.2%	8	22
Schroder Asian Total Return	4.7%	2	10
Worldwide Healthcare	4.7%	9	38

Appendix L · Data coverage by year (all panel rows incl. VCT lines)

YEAR	SECURITIES	ROWS	PRICE COV.	RETURN COV.
2005.0	3	5	100.0%	0.0%
2006.0	52	85	100.0%	17.6%
2007.0	443	4658	68.0%	59.9%
2008.0	472	5133	59.8%	56.7%
2009.0	472	5126	55.4%	52.7%
2010.0	472	5112	52.5%	50.3%
2011.0	461	5108	50.9%	49.0%
2012.0	449	5029	53.1%	47.5%
2013.0	441	4932	49.6%	48.1%
2014.0	449	5029	50.4%	49.2%
2015.0	445	5049	50.6%	49.2%
2016.0	441	4944	51.3%	49.8%
2017.0	441	4889	51.2%	49.4%
2018.0	442	4877	50.8%	49.9%
2019.0	427	4904	49.4%	48.7%
2020.0	423	4876	49.6%	48.7%
2021.0	429	4788	48.7%	47.9%
2022.0	402	4685	48.6%	47.7%
2023.0	388	4495	48.0%	47.0%
2024.0	368	4170	47.5%	46.9%
2025.0	327	3716	48.6%	47.8%
2026.0	292	1988	49.5%	48.6%

Eligible conventional universe coverage is near-complete; VCT rows lack monthly prices by design.

APPENDIX M · Complete chart gallery

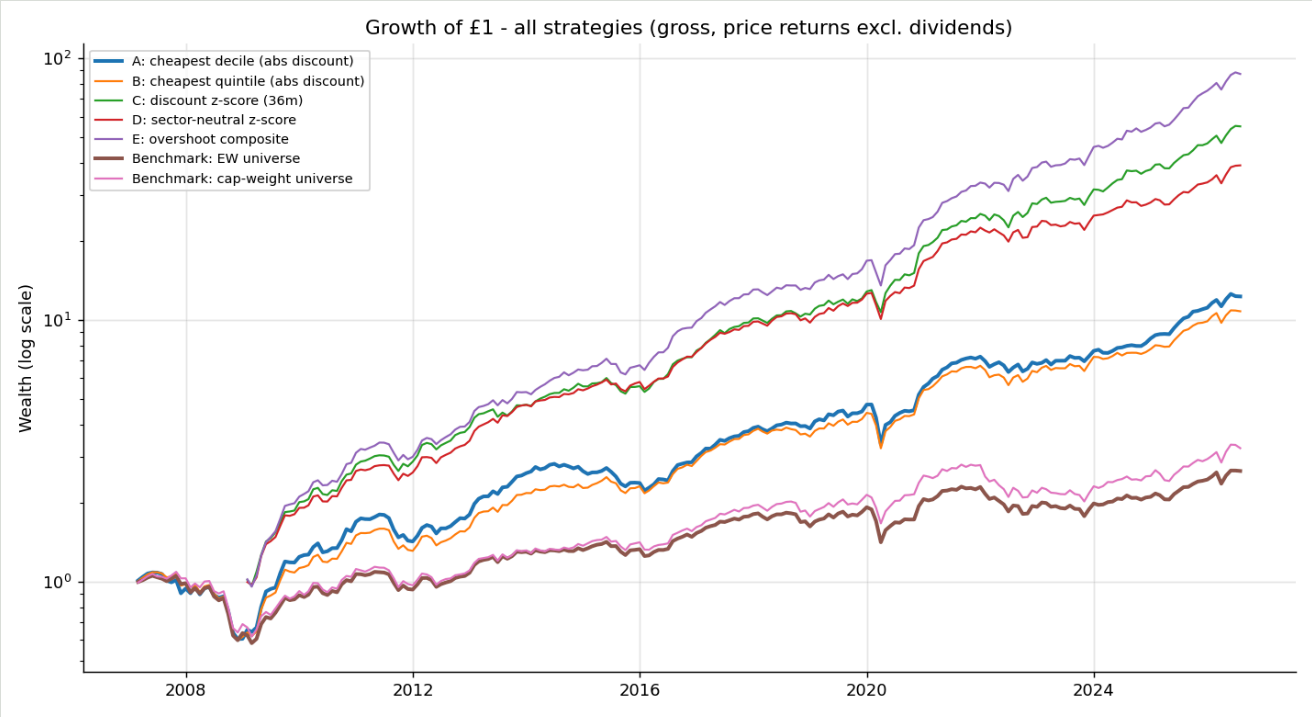

Chart 01 — Growth of £1 — all strategies (gross)

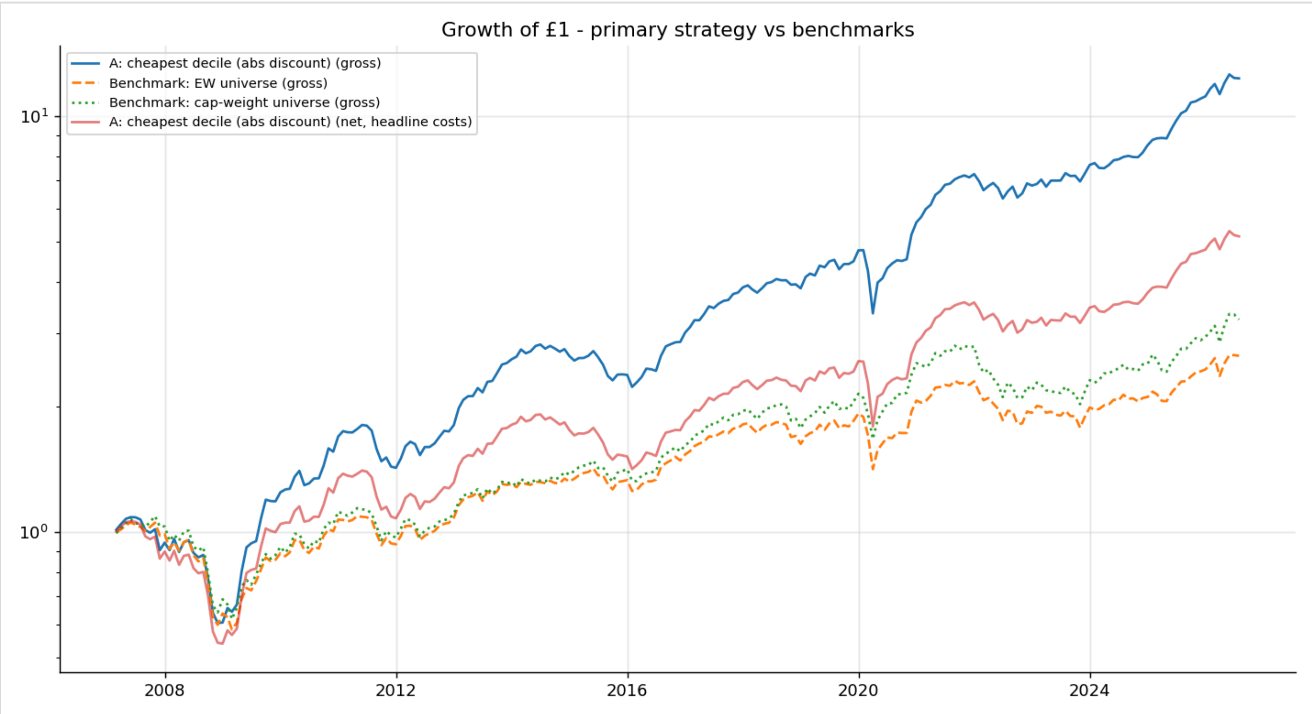

Chart 02 — Primary strategy vs benchmarks

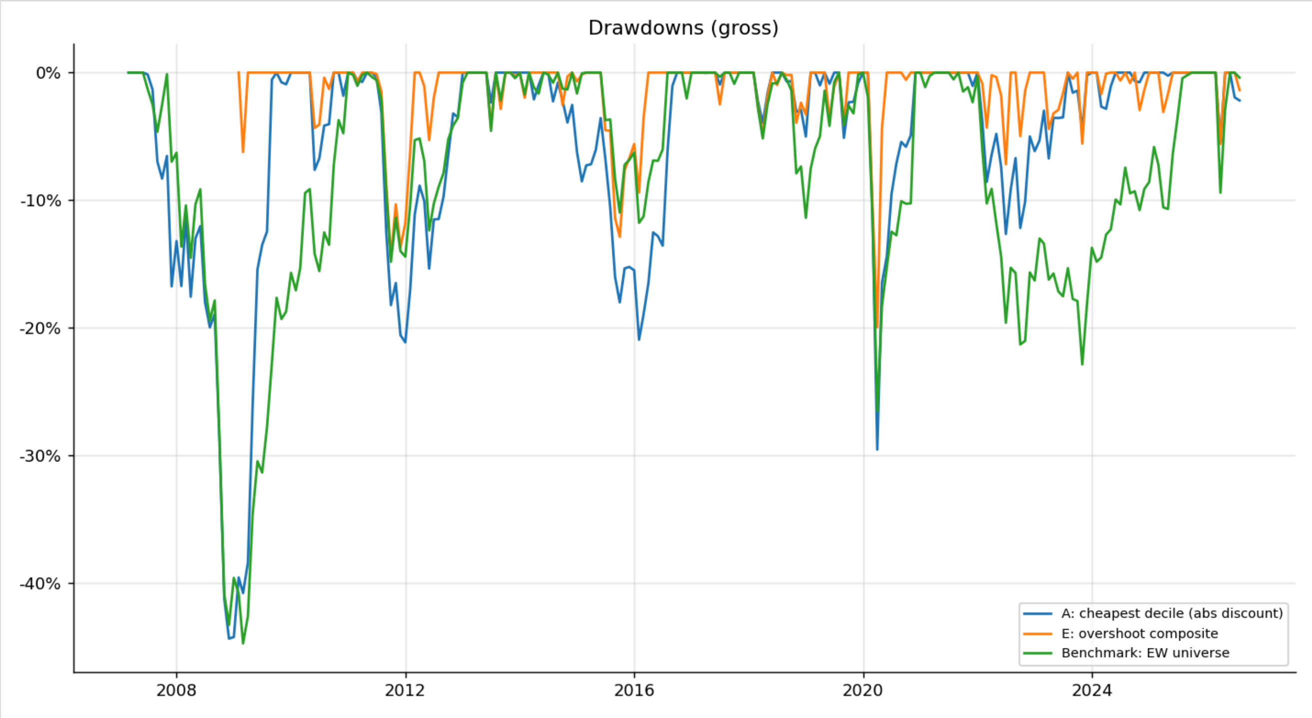

Chart 03 — Drawdowns

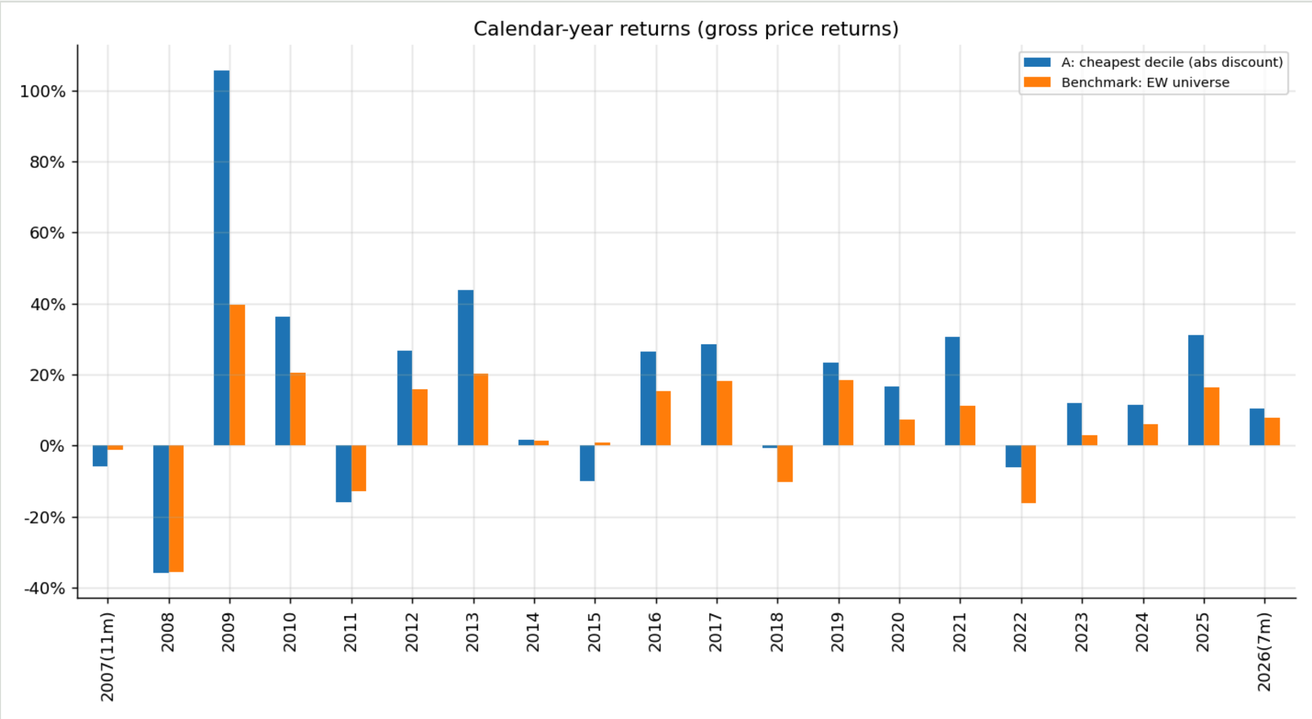

Chart 04 — Calendar-year returns

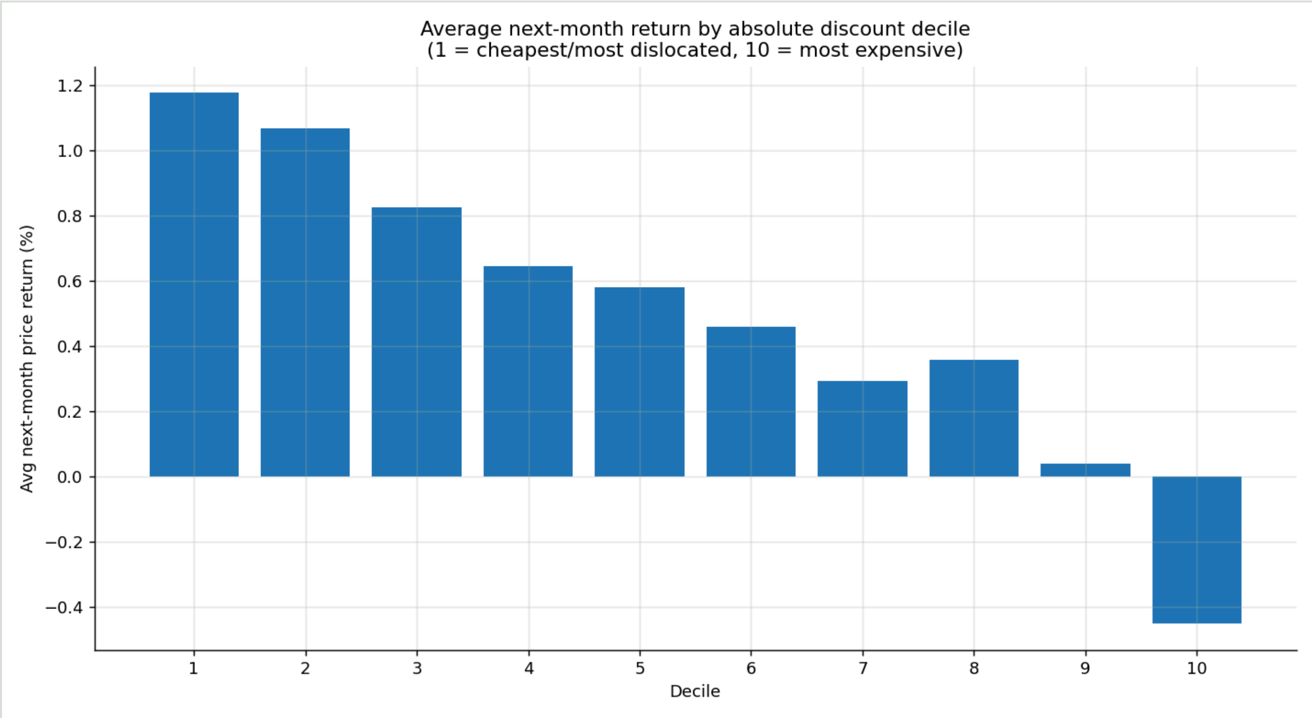

Chart 05 — Discount-decile forward returns

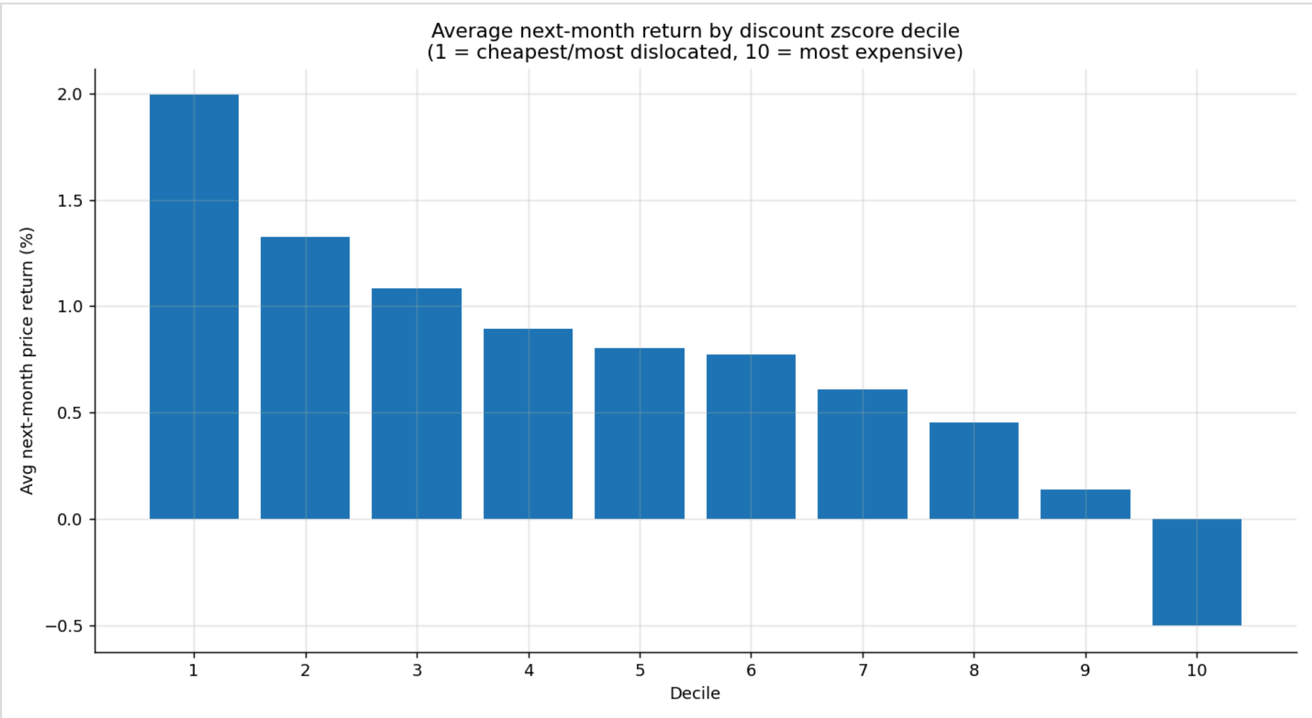

Chart 06 — Z-score-decile forward returns

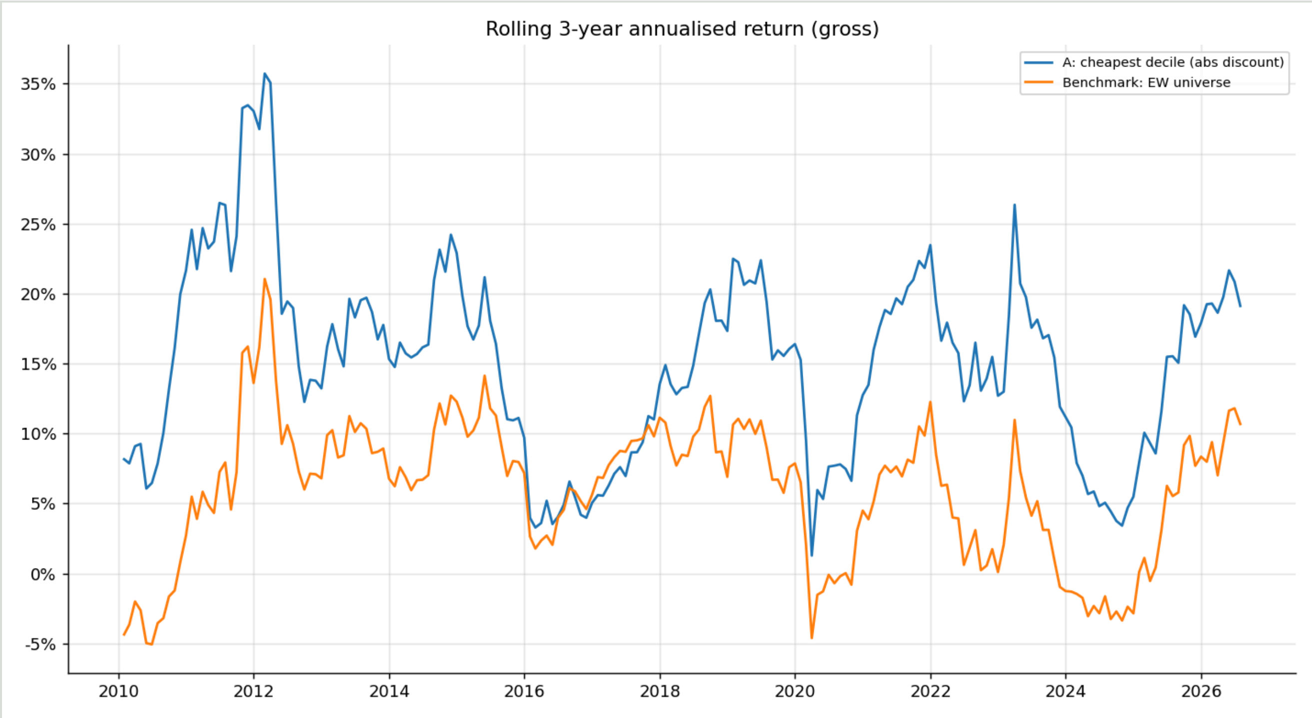

Chart 07 — Rolling 3-year returns

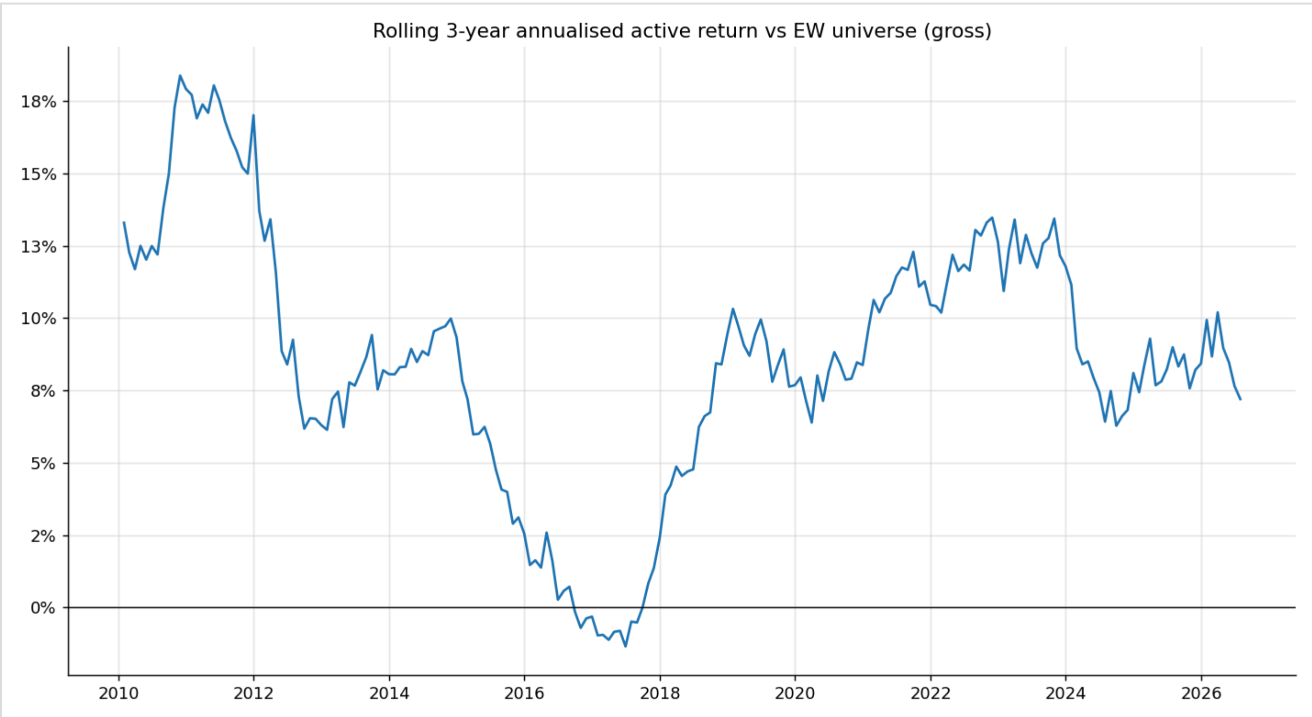

Chart 08 — Rolling 3-year active return

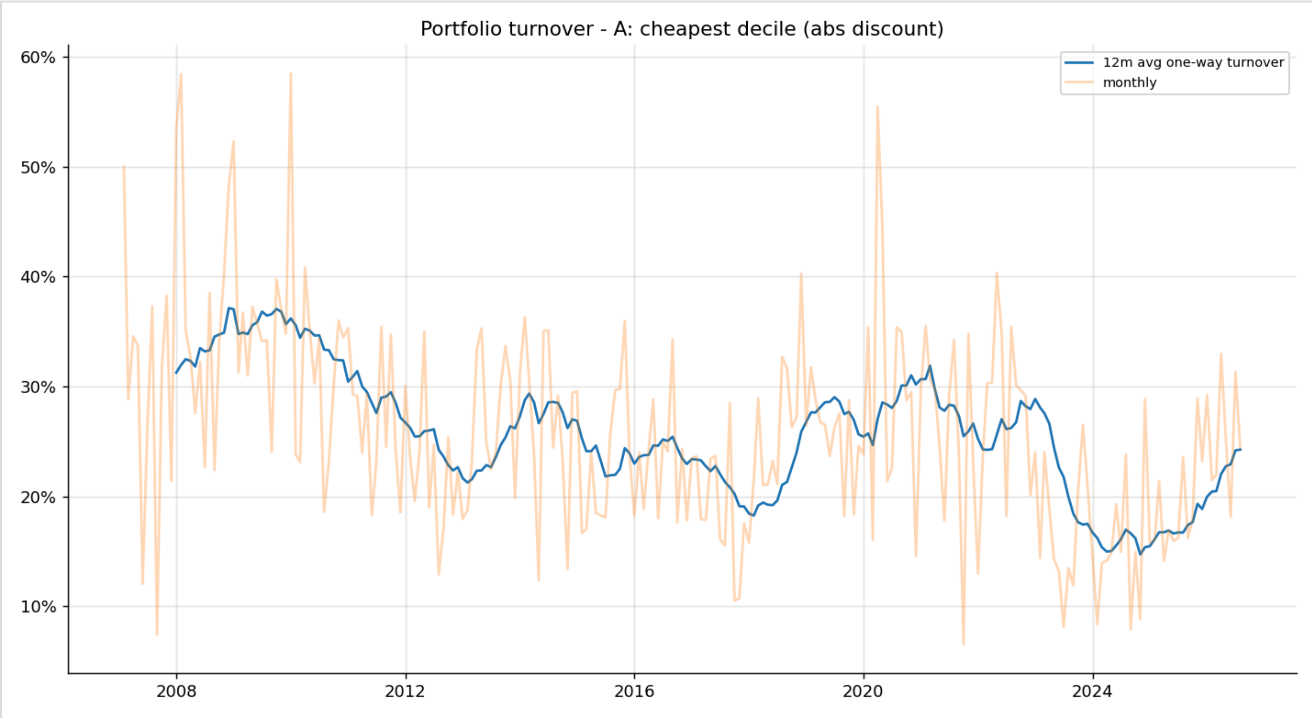

Chart 09 — Portfolio turnover

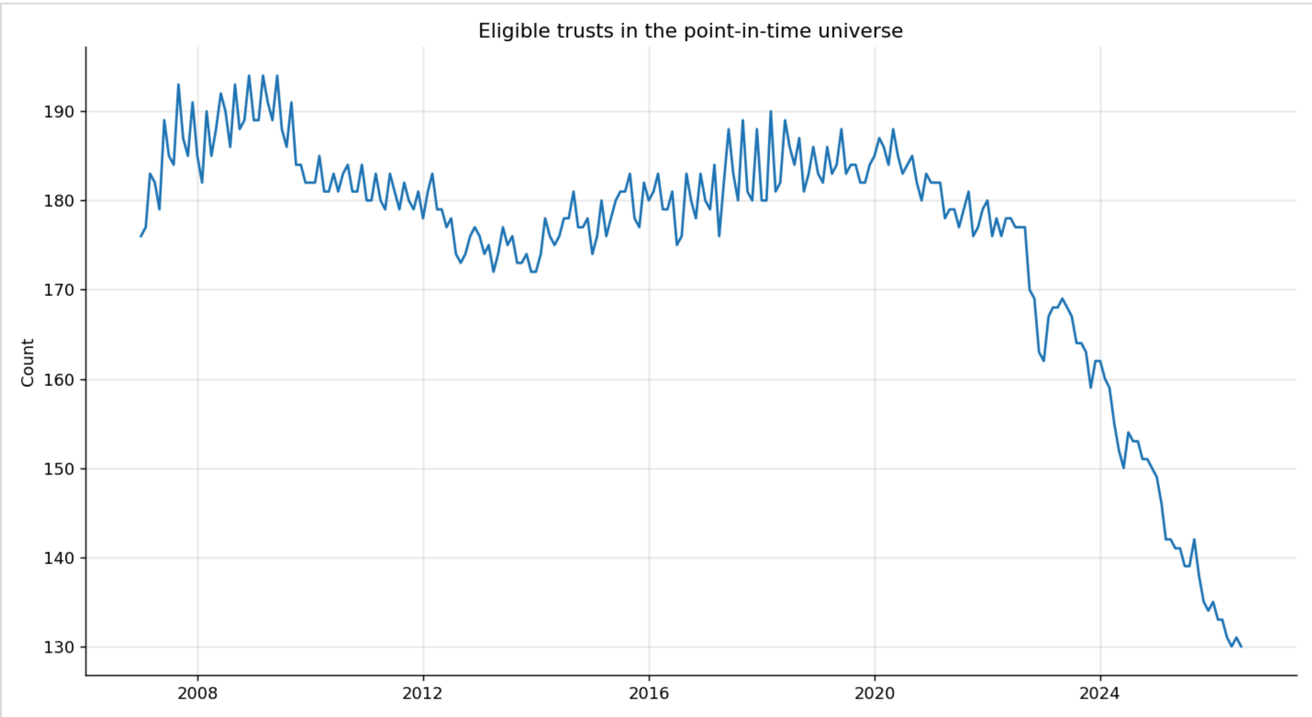

Chart 10 — Eligible universe size

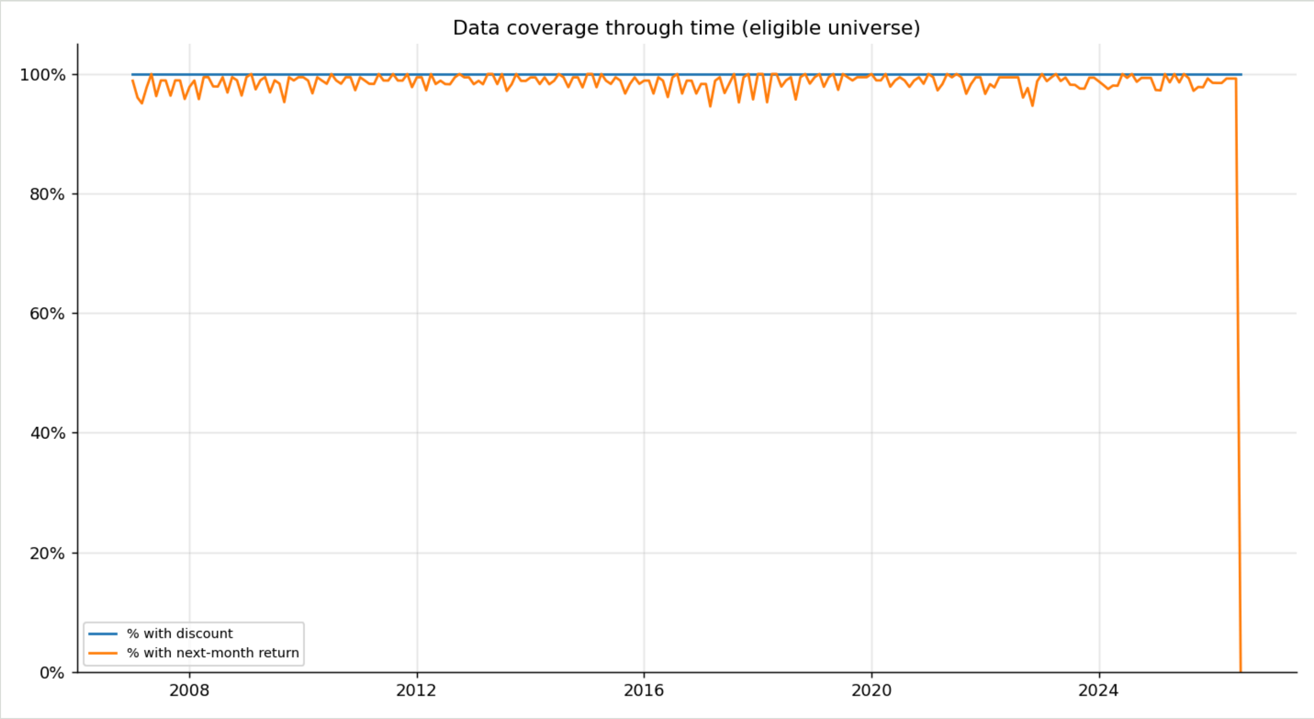

Chart 11 — Data coverage

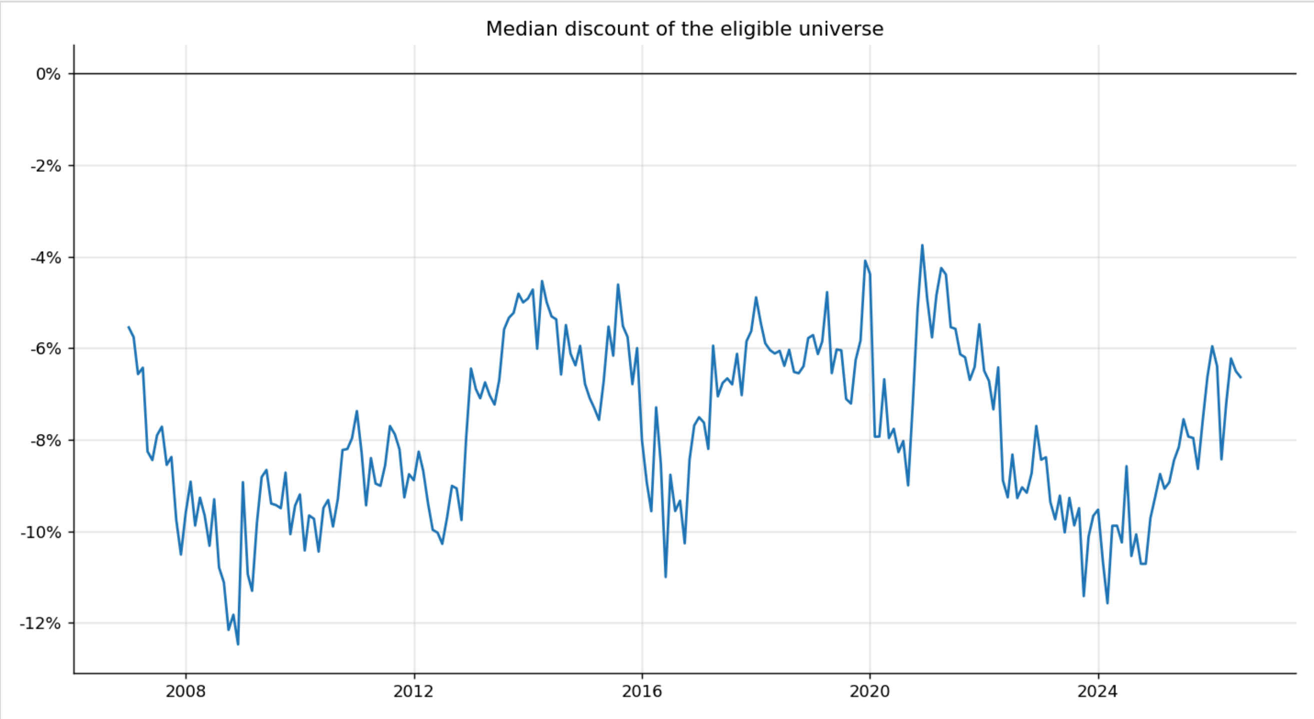

Chart 12 — Median universe discount

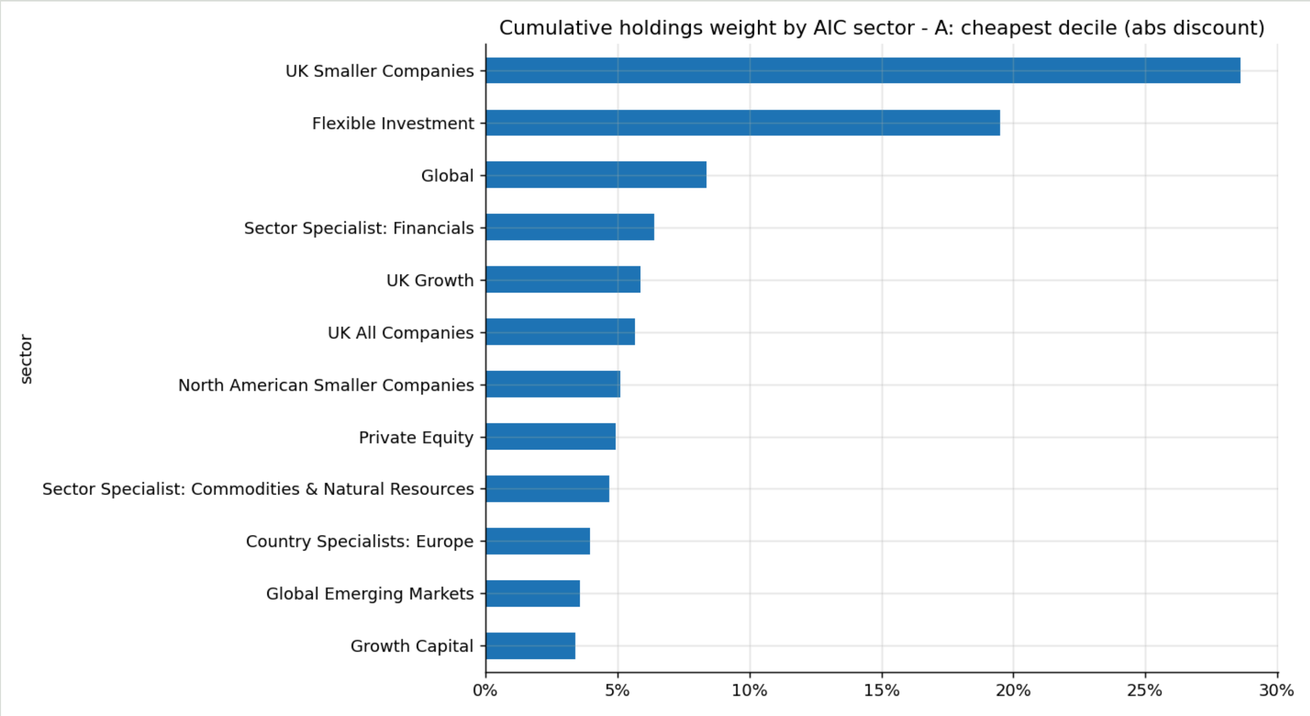

Chart 13 — Holdings by AIC sector

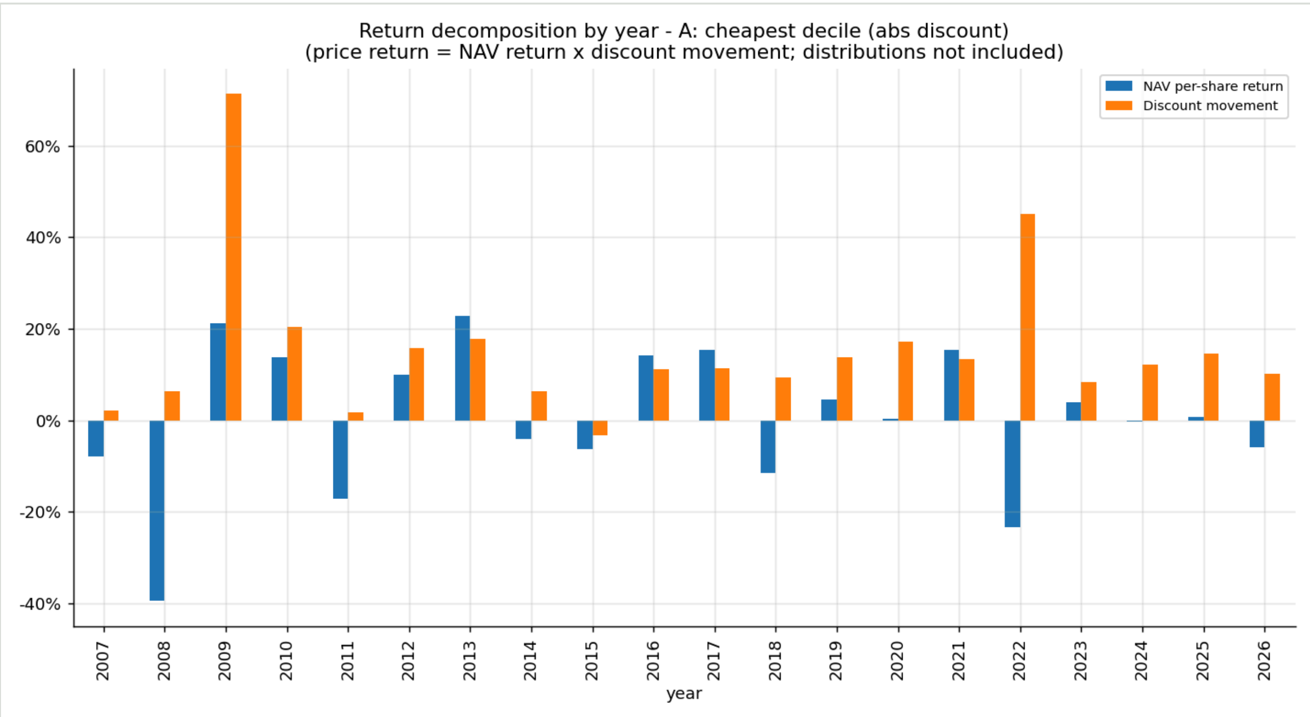

Chart 14 — Return decomposition by year

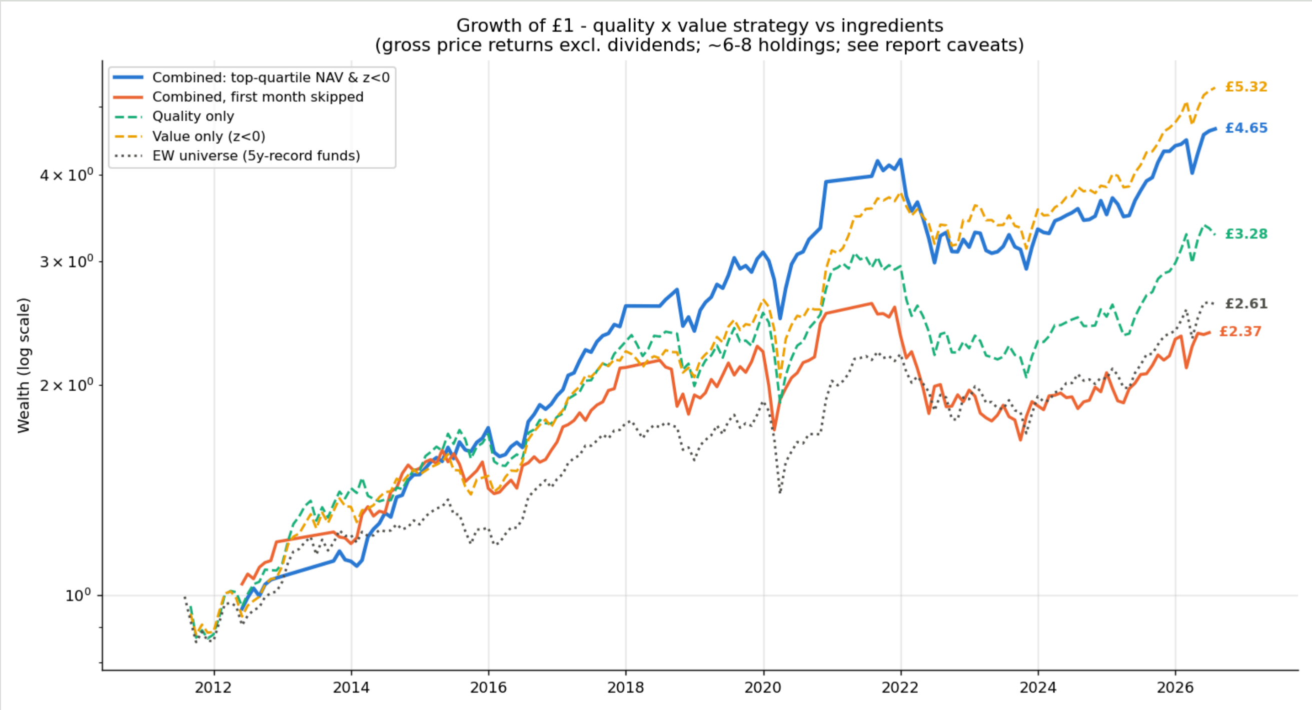

Chart 15 — Quality × value growth (corrected)

Historical backtests are research tools, not investment advice. No missing financial data has been synthetically filled. Sources: Association of Investment Companies data archive (MIR, industry overviews, corporate activity, 2007–2026); Investegate regulatory news archive. Repository: hdcapital/cef-research — the entire study rebuilds via python -m uk_cef.cli run-all.